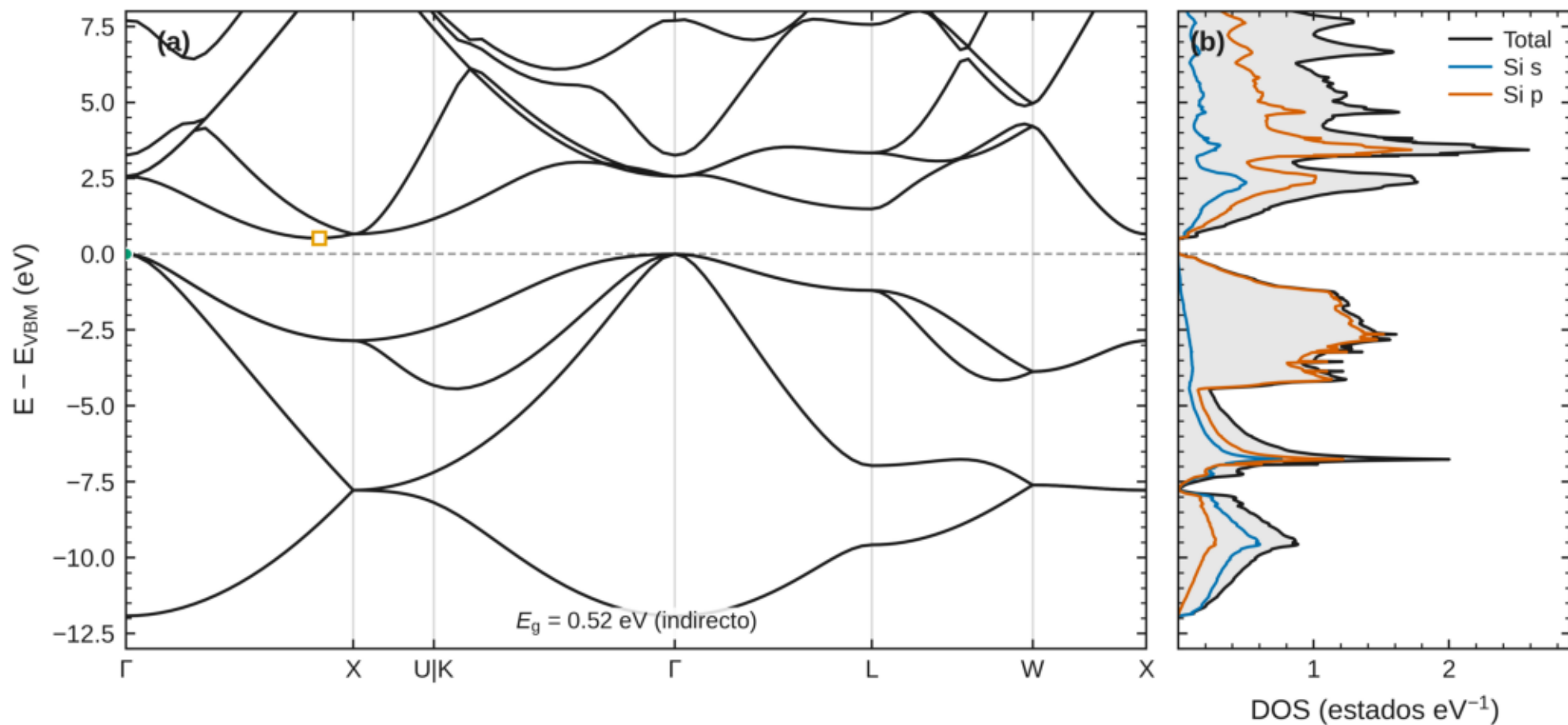


# Bands and DOS · silicon

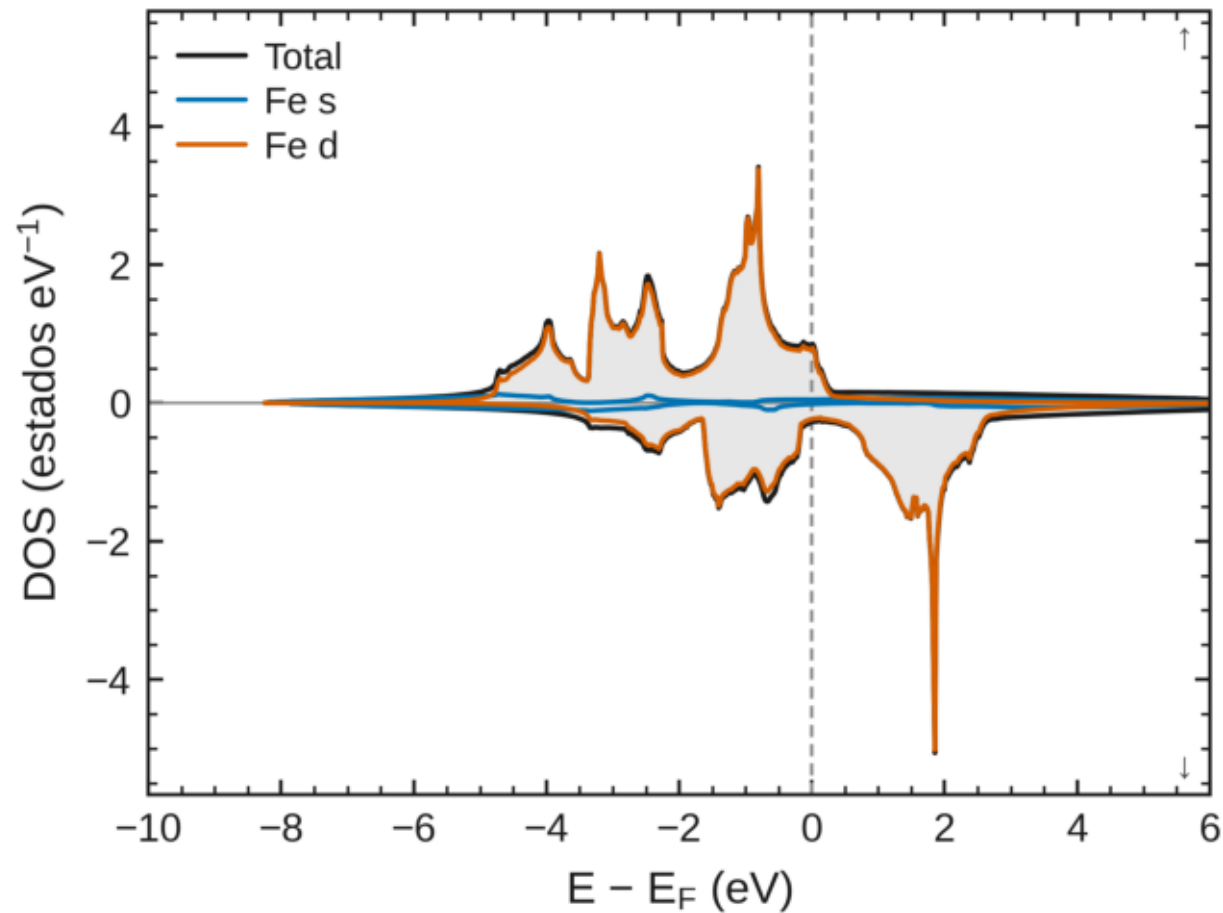


LDA example; calculated gap 0.52 eV. This is not the experimental gap.

[Existing example · conditions and source files](#)

Jorge Enrique González Sevilla · GPL-3.0-or-later · No new simulations for this gallery.

# Magnetism · bcc iron

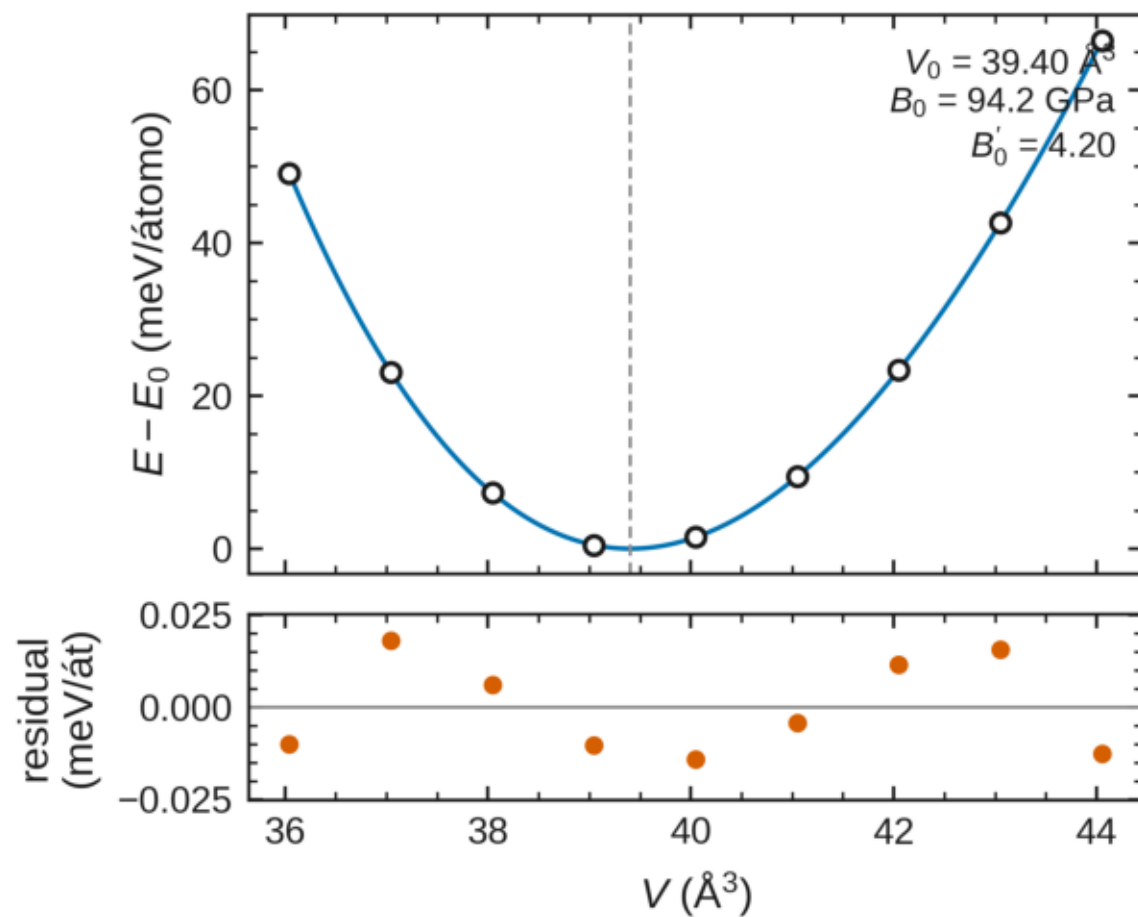


Spin-resolved DOS of a ferromagnetic example. Opposite channels follow the plotting convention.

[Existing example · conditions and source files](#)

Jorge Enrique González Sevilla · GPL-3.0-or-later · No new simulations for this gallery.

# Equation of state · silicon

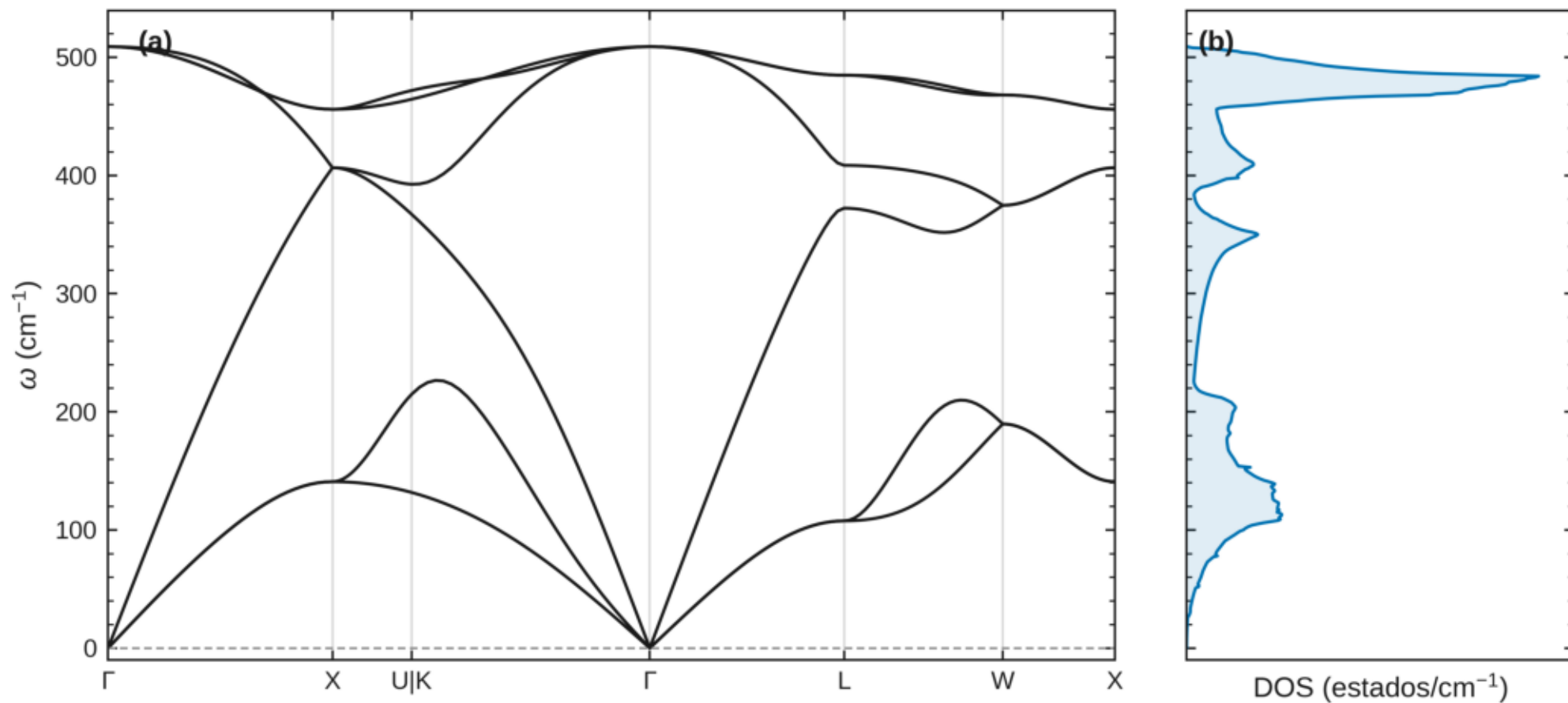


Energy versus volume and Birch-Murnaghan fit residuals. LDA example.

[Existing example · conditions and source files](#)

Jorge Enrique González Sevilla · GPL-3.0-or-later · No new simulations for this gallery.

# Lattice vibrations · silicon

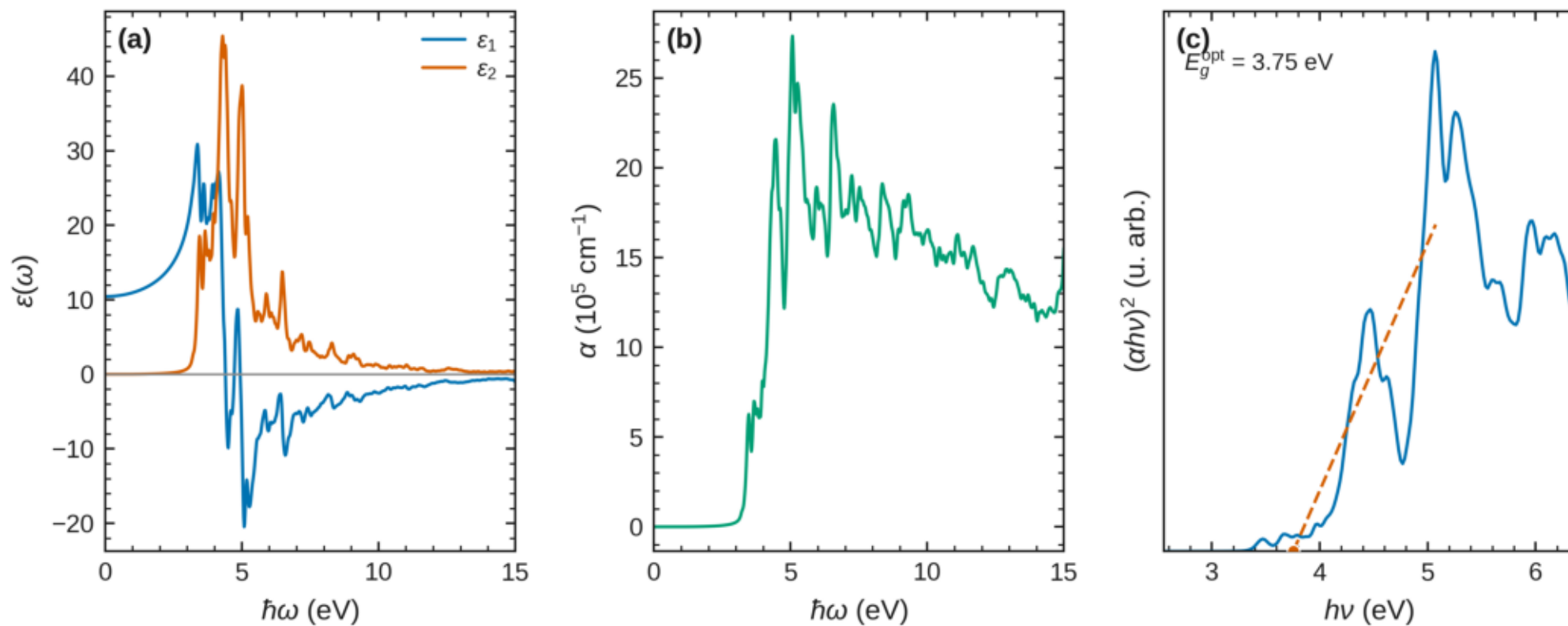


Phonon dispersion and DOS. LDA example,  $2 \times 2 \times 2$  q mesh; not a convergence claim for other materials.

[Existing example · conditions and source files](#)

Jorge Enrique González Sevilla · GPL-3.0-or-later · No new simulations for this gallery.

# Optical response · silicon



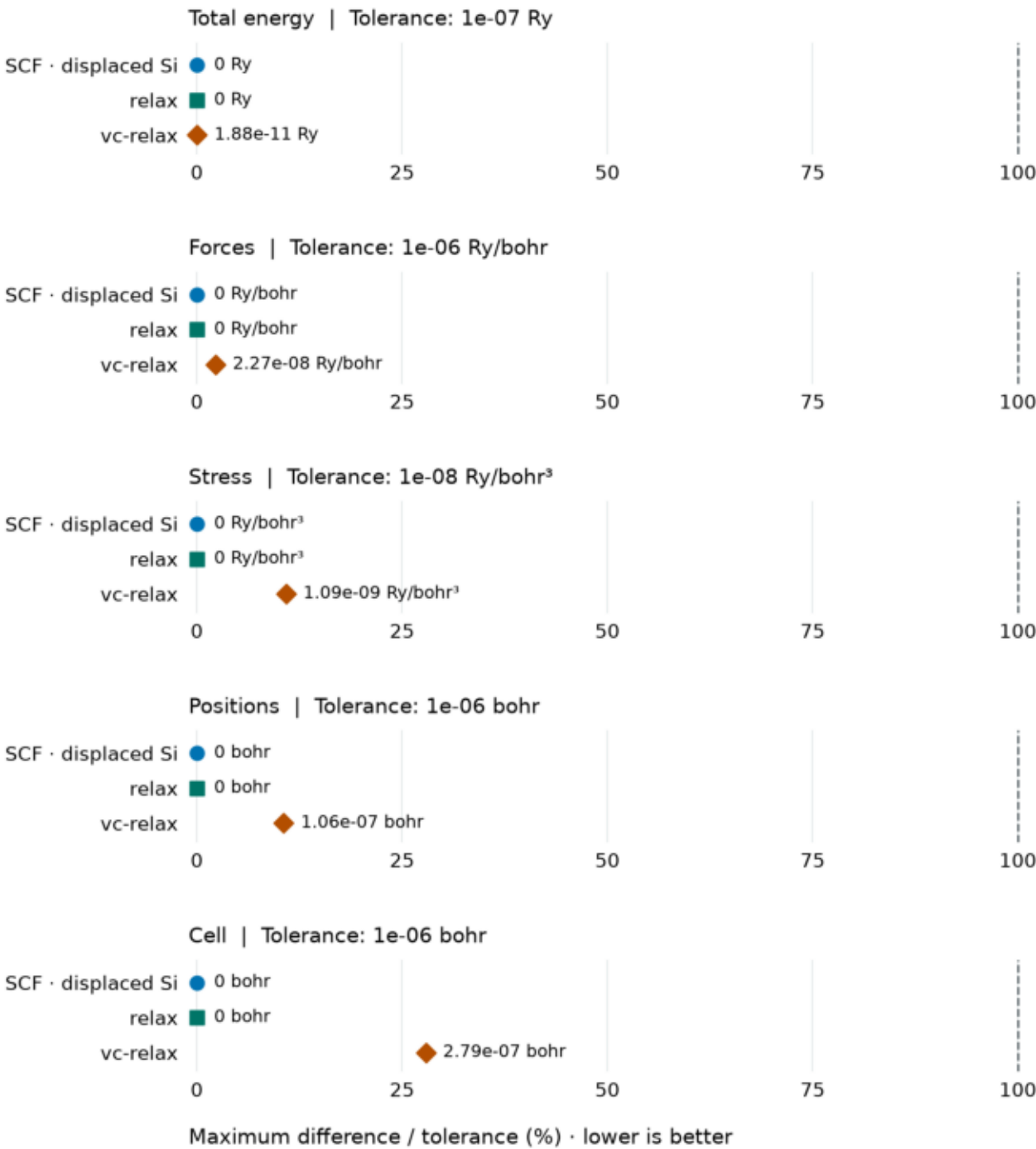
Example optical functions with a 0.65 eV scissor correction. The displayed optical fit is not the indirect electronic gap.

[Existing example · conditions and source files](#)

# Olla-DFT 1.2.0 · local recovery validation

## Difference from the continuous execution

Three pairs of serial QE 7.4 jobs; same inputs and environment within each pair.



Labels show absolute differences, not uncertainties. 100% = acceptance limit.  
0 means identical at XML precision. One pair per case; no statistical inference.  
Simulated local interruptions; recovery after physical power loss or disk loss was not measured.